

**UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Advanced Subsidiary Level and GCE Advanced Level**

**MARK SCHEME for the October/November 2006 question paper**

**9702 PHYSICS**

**9702/06**

Paper 6 (Options), maximum raw mark 40

This mark scheme is published as an aid to teachers and students, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

The grade thresholds for various grades are published in the report on the examination for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses.

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### Option A – Astrophysics and Cosmology

- 1 (a)** gaseous/rocky/icy/approx. spherical object that orbits the Sun / a star B1 B1 [2]
- (b)** Venus must have passed between Sun and Earth 1 AU is (mean) distance between Earth and Sun B1 B1 [2]
- 2 (a)** (light of a particular wavelength as observed) when source is moving away (from observer) has a longer wavelength than when source is stationary (with respect to observer) M1 A1 B1 [3]
- (b)** (extent of) redshift depends on  $v/c$  can only be observed when  $v$  is significant when compared to  $c$  B1 B1 [2]
- 3 (a)**  $v$  is speed of separation of (any two) galaxies  $d$  is the separation of the galaxies (max 1 mark if refers to Earth) B1 B1 [2]
- (b)**  $1 \text{ Mpc} = 3.09 \times 10^{19} \text{ km}$  (allow  $3.0 \rightarrow 3.2$ )  
 $\text{age} = 1 / H_0$   
 $\text{age} = (3.09 \times 10^{19}) / 60$   
 $= 5.2 \times 10^{17} \text{ s}$  C1 C1 A1 [3]
- 4 (a)** e.g. dark matter does not emit light dark matter does not reflect light (any two sensible suggestions, 1 each) B2 [2]
- (b)** e.g. estimate of mass unreliable because there are neutrinos e.g. do not know extent of Universe due to redshift / intensity of light (any sensible suggestion (M1) with reason (A1)) M1 A1 M1 A1 [4]

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### Option F – The Physics of Fluids

- 5 (a) symmetrical pattern 'above' and 'below'  
lines closest together at widest part of object  
smooth lines tending towards initial separation  
B1  
B1  
B1 [3]
- (b) *either* separation of lines is not constant *or* path lengths differ  
B1 [1]
- 6 (a) centre of buoyancy is above the centre of mass  
(if displaced sideways) weight and upthrust provide couple to keep tube upright  
(do not allow argument in terms of metacentre)  
B1  
B1 [2]
- (b) *either* force on base =  $L\rho g \times A$  *or* weight of liquid displaced =  $\rho LA g$   
this equals weight  $Mg$  this equals weight  $Mg$   
hence  $L = M/A\rho$  hence  $L = M/A\rho$   
M1  
A1  
A0 [2]
- (c)  $M/A = L\rho = \text{constant}$   
new length =  $12.1 \times (0.99/1.11) = 10.8 \text{ cm}$   
change in length = 1.3 cm  
C1  
C1  
A1 [3]
- 7 (a) (apparent) weight acts downwards  
drag force acts upwards  
resultant force = weight -  $k v$  OR drag  $\propto v$   
as speed increases, resultant force / acceleration becomes less  
(so) speed increases to a constant value  
B1  
B1  
B1  
B1  
B1 [5]
- (b) fluid is dragged along by the surface of the (spinning) sphere  
on one 'side' speed of fluid is greater than on other  
this difference in speed creates a pressure difference / difference in drag / turbulence  
so sphere moves sideways (in direction of lower pressure)  
B1  
M1  
A1  
A1 [4]

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### Option M – Medical Physics

- 8 (a) pulse of ultrasound B1  
reflected at boundary (between any two media) B1  
reflected pulse detected by (piezo-electric) crystal B1  
signal from crystal amplified / processed and displayed B1 [4]
- (b) crystals are at different orientations B1  
signals from all crystals are combined B1  
to build up a (2D) image B1 [3]
- 9 (a) (i) process by which objects at different distances from the eye M1  
are brought to a focus (on the retina) A1 [2]  
(ii) ciliary muscles alter shape of lens B1  
this alters the power/focal length of the lens B1 [2]
- (b) pupil varies in diameter C1  
power (intensity) admitted is proportional to diameter<sup>2</sup> B1  
*either* variation of diameter is small / small factor  
*or* variation of light intensity is large / (very) large factor B1 [3]
- 10 (a)  $IL = 10 \lg(I/I_0)$   
 $= 10 \lg(\{1.6 \times 10^{-10}\} / \{1 \times 10^{-12}\})$   
 $= 22 \text{ dB}$  C1  
range is from 100 Hz A1  
to 10 kHz B1 B1 [4]
- (b) e.g. threshold intensity rises  
upper frequency (limit) decreases  
lower frequency (limit) rises  
(*any two suggestions, 1 each, max 2*)  
(*allow 1 mark for 'line closes up' / smaller frequency range*) B2 [2]

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### Option P – Environmental Physics

- 11 (a)**
- (i) slows down neutrons  
to enable further fission reactions  
M1  
A1 [2]
  - (ii) absorbs neutrons  
to control rate of reaction / power  
M1  
A1 [2]
  - (iii) acts as a biological shield  
B1  
maintains coolant around the core / containment vessel  
B1 [2]
- (b)** kinetic energy of fission fragments  
causes heating of the core / fuel rods  
B1  
this thermal energy is carried away by the coolant  
B1  
B1 [3]
- (c)** e.g. *either* minimal / no release of CO<sub>2</sub> into the atmosphere  
or minimal / no release of gases causing global warming  
no huge storage areas required at the power station  
maintenance possible whilst on full load  
(any two suggestions, 1 each, max 2)  
B2 [2]
- 12 (a)** incident power =  $960 \times 2.5 \times 10^{-4} = 0.24 \text{ W}$   
efficiency =  $(30 \times 10^{-3}) / 0.24 = 0.13$   
C1  
A1 [2]
- (b)**
- (i) large (surface) area required  
B1
  - (ii) connect many cells in series for higher voltage  
B1  
connect many cells in parallel for larger current  
B1 [3]
- 13 (a)** 30% delivered to motor  
cost =  $5.4 \times (100/30) \times 5$   
= 90 cents  
A1 [2]  
(allow 1 mark for answer 100 cents)
- (b)** (for both,) there is a need to heat water / for heat energy  
this energy provided from 'production losses' (so reducing overall costs)  
B1  
B1 [2]

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### Option T – Telecommunications

- 14 (a)** satellite with orbit having period 24 hours  
orbits above the Equator  
from west to east / orbits in same sense  
B1  
B1  
B1 **[3]**
- (b)**  $\text{loss} = 10 \lg(P_2 / P_1)$   
 $-170 = 10 \lg(P_2 / 2400)$   
 $P_2 = 2.4 \times 10^{-14} \text{ W}$   
C1  
C1  
A1 **[3]**
- (c)** amplified otherwise power too low to be picked up on Earth  
*either* frequency changed to prevent swamping / interference of signal received (from Earth)  
*or* prevent feedback  
B1  
B1 **[2]**
- 15 (a)** variations in either amplitude or frequency of a wave  
*either* in synchrony with displacement of information signal  
*or* in order to carry information on the wave  
B1  
B1 **[2]**
- (b) (i)** 9 kHz  
**(ii)** LW frequency range is 30 kHz  $\rightarrow$  300 kHz  
number =  $270 / 9$   
= 30  
B1 **[1]**  
C1  
C1  
A1 **[3]**
- (c)** sketch: carrier frequency as vertical line and two sidebands  
reasonable symmetry  
sideband indicating approx. 4500 Hz range  
(*if sidebands shown as vertical lines, allow max. 1 mark*)  
M1  
A1  
B1 **[3]**
- 16 (a)** e.g. link between house and exchange for a telephone  
(*any one suggestion, 1 mark*)  
B1 **[1]**
- (b)** e.g. greater bandwidth  
less noise  
less attenuation  
(*any two suggestions, 1 each, max 2*)  
B2 **[2]**